

WEEK PRACTICAL – 01

1.id:

```
navaneeth@Navaneeth:~$ id
uid=1000(navaneeth) gid=1000(navaneeth) groups=1000(navaneeth),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
navaneeth@Navaneeth:~$ ;5~
```

Report: After running the id command, I could see my user details. It showed my username as Navaneeth, along with the user ID, group ID, and the groups I belong to like sudo and plugdev.

2.env:

```
_DISTRO_NAME=Ubuntu
ME=Navaneeth
D=/root
GNAME=root
TD_SHOWN=update-motd
ME=/root
NG=C.UTF-8
_INTEROPP=/run/WSL/392_interop
_COLORS=rs=0:di=01;34:ln=01:36:mh=00:pi=40;33:so=01:35:do=01;35:bd=40;33:0
:ex=01;32:*:tao=01;31*:tgz=01;31*:arc=01;31*:arj=01;31*:laz=01;31*:lh
:7z=01;31*:zip=01;31*:z=01;31*:dz=01;31*:gz=01;31*:lrz=01;31*:lz=01
:l=31*:tbz2=01;31*:tz=01;31*:deb=01;31*:rpm=01;31*:jar=01;31*:war=01
01;31*:7z=01;31*:xzz=01;31*:cab=01;31*:wim=01;31*:swm=01;31*:dwm=01;
if=01;35*:bmp=01;35*:pbm=01;35*:pgm=01;35*:ppm=01;35*:tga=01;35*:xbm
:mng=01;35*:pcx=01;35*:mov=01;35*:mpg=01;35*:mpeg=01;35*:m2v=01;35*:
35*:vob=01;35*:qt=01;35*:nuv=01;35*:wmv=01;35*:asf=01;35*:rm=01;35*:
*:xcf=01;35*:xwd=01;35*:yuv=01;35*:cgy=01;35*:emf=01;35*:ogv=01;35*:
*:mka=00;36*:mp3=00;36*:mpc=00;36*:ogg=00;36*:ra=00;36*:wav=00;36*:
download=00;90*:dpkg-dist=00;90*:dpkg-new=00;90*:dpkg-old=00;90*:dpkg-
ig=00;90*:rpmsave=00;90*:swp=00;90*:tmp=00;90*:ucf-dist=00;90*:ucf-ne
YLAND_DISPLAY=wayland-0
SSCLOSE=/usr/bin/lesspipe %s %s
RM=xterm-256color
SSOPEN=| /usr/bin/lesspipe %s
ER=root
SPLAY=:0
LVL=1
G_RUNTIME_DIR=/run/user/0
LENV=
G_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
TH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games
avapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt
t/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/windows
t/c/Program Files/nodejs:/mnt/c/Users/saicharan_6wg39hp/AppData/Local/Microsc
/bin:/mnt/c/Program Files/JetBrains/PyCharm 2026.1/bin:/mnt/c/Users/navanes
reddi/AppData/Local/Office/12.1.0.274586/clitool:/snap/bin
US_SESSION_BUS_ADDRESS=unix:path=/run/user/0/bus
STTYPE=x86_64
LSE_SERVER=unix:/mnt/wslg/PulseServer
/usr/bin/printenv
navaneeth@Navaneeth:~$
```

Report: When I executed the env command, many environment variables were displayed. I observed variables like HOME=/home/navaneeth, USER=navaneeth, PWD=/home/navaneeth/OSSP, and PATH, which contain information about the system environment.

```
35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fl=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*=00;90:*=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpkg-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:"
declare -x NAME="SRIVALLI"
declare -x OLDPWD
declare -x PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Users/sriva_6wg39hp/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/sriva_6wg39hp/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/JetBrains/PyCharm 2026.1/bin:/mnt/c/Users/sriva_6wg39hp/AppData/Roaming/npm:/mnt/c/Users/sriva_6wg39hp/AppData/Local/Kingsoft/WPS Office/12.1.0.27458/clitool:/snap/bin"
declare -x PULSE_SERVER="unix:/mnt/wslg/PulseServer"
declare -x PWD="/root"
declare -x SHELL="/bin/bash"
declare -x SHLVL="1"
declare -x TERM="xterm-256color"
declare -x USER="root"
declare -x WAYLAND_DISPLAY="wayland-0"
declare -x WSL2_GUI_APPS_ENABLED="1"
declare -x WSLENV=""
declare -x WSL_DISTRO_NAME="Ubuntu"
declare -x WSL_INTEROP="/run/WSL/643_interop"
declare -x XDG_DATA_DIRS="/usr/local/share:/usr/share:/var/lib/snapd/desktop"
declare -x XDG_RUNTIME_DIR="/run/user/0"
```

4. Alias:

```
navaneeth@Navaneeth:~$ alias
alias alert='notify-send --urgency=low -i "${([ $? = 0 ]) && echo terminal || echo error}" "${history|tail -n1}sed -e '\''s/\s*[0-9]+\s*//;s/[/;|]\s*alert$/\s*\s*\s*'\''"'
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias ls='ls -CF'
alias la='ls -A'
alias ll='ls -lF'
alias ls='ls --color=auto'
navaneeth@Navaneeth:~$
```

TASK-1 Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to 3 another. Explain how control transitions between user Space and kernel space during execution



navaneeth@Navaneeth: ~



GNU nano 8.7.1

```
#include <unistd.h>
#include <stdlib.h>
#include <stdio.h>
#include <fcntl.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    int source, destination;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    source = open(argv[1], O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytes_read = read(source, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(destination, buffer, bytes_read);
        if (bytes_written != bytes_read)
        {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }

    if (bytes_read < 0)
    {
        perror("Error reading source file");
    }

    close(source);
    close(destination);

    return 0;
}
```